

A Reflective Journey: Navigating Your Cumulative Experience at Iowa State University

Reflecting on my years as a student at Iowa State University, I can see that my education in electrical engineering has been more than just coursework and a degree. At the time of starting my university life, I viewed the role of an engineer as a person who has to solve technical problems using math and sciences. In pursuing my major in Electrical Engineering, I realized that being an engineer involves much more than that. An engineer is supposed to solve complex problems, communicate with people from various fields, be a good team player, consider the impacts of his decisions on other people, and constantly learn because of advancements in technology.

The first major shift for me is that engineering problems should be considered from the perspective of a bigger picture. In college, it is often believed that there is only one right answer. In practice, however, things are different. It is through my electrical engineering classes that I have come to understand that just because something works does not mean that it is a good solution. Engineers need to take into consideration such aspects as cost, reliability, safety, efficiency, ecological impacts, and even the people that would benefit from the engineering project. The importance of it became clear as I started to study more and more about power systems. For instance, when studying power system analysis and economics of electric power planning, I start thinking not only about the behavior of the system electrically but also about the economics of its operation. Generating and transmitting electricity is related to utilities, their customers, communities, and even the environment.

In the course of my college education, I have come to understand a lot more about engineering problem solving. Initially, my approach would be to find the relevant equation and try to solve it. However, with increased difficulty of class problems, this became inadequate. This necessitated that I first gain an understanding of the system, know the available data, identify the objective, and then select the appropriate solution approach. Each class provided its own challenge. The circuit courses taught me how to analyze the electrical behavior using fundamental principles, Signals and Systems made me think in mathematical domains such as time, frequency, and Laplace domains, and the integrated circuit design presented yet another form of problem whereby the selection made at the transistor level impacts the circuit as a whole. These were all aimed at encouraging power thinking which is necessary to deal with complex systems in power courses.

Project work has been particularly helpful in making this transition. For instance, I created a digital potentiometer design on Cadence Virtuoso. To do so, it wasn't enough for me to know how particular transistors behave individually. It required me to understand how all the components such as switches, resistor chains, and a decoder would work as a whole system.

While comparing various designs, I realized how one was more accurate, but the other was more stable. There is also another experience when my team created a robot that would use sensors and its programmed behaviors to detect and engage with any object. It made me realize that the initial idea doesn't always have to be the result.

This particular lesson holds even greater value to me through my Senior Design work. The team I am working with designs an eddy current probe that will measure conductivity. It is not like most homework assignments when there is a page in the textbook that guides us on how to construct a particular device. We need to study the basic theories, design the circuit, calculate the measurements, find the components, perform experiments in the lab, and modify our design based on the results obtained. Even during the preliminary discussion for our project, it was apparent how critical it is for us to have members on the team who bring different expertise in order to examine the problem from different perspectives. Some of our team members understand the concepts of integrated circuitry, RF concepts, material science, and power systems.

Another aspect of development is acquiring information, not just in the classroom environment. I realized at college that lectures cannot provide engineers with all necessary information. I used textbooks, component manuals, technical literature, software, peers, professors, and experts for technical assignments and tasks. The experience gained during my internship at Electrical Power Products (EP2) confirmed this point. As an intern in the Quality Engineering department, I encountered equipment and procedures that I was not familiar with. It included instruments such as Omicron CMC 356, protective relays, testing equipment, and electrical tests themselves. It was impossible to get all necessary information only through lectures. I had to learn everything by analyzing existing documents, observing other testers, and asking questions.

As one of my main responsibilities in the internship, I was asked to write and enhance the existing work instructions for manufacturing and testing processes. The experience taught me something important about engineering communication. Prior to the internship, I was mainly viewing engineering communication as writing a report or presenting a project. At EP², I realized that poor technical writing could have a direct influence on the performance of a certain task. A work instruction had to be technically accurate and easily comprehensible for an operator of the procedure. After receiving feedback from testers, quality engineers, and management, I revised the instructions accordingly. This process convinced me that feedback is a natural part of engineering, not a failure. In addition, it is confirmed that knowledge can be acquired from various people. The person who performs the same procedure daily can point out a problem that is not visible on the drawing or in the document.

The internship taught me how learning occurs outside the classroom setting as well. It allowed me to understand how engineering interacts with manufacturing, quality, testing, and business processes. It made me realize that engineers and technicians have to consider safety, repeatability, record keeping, timetables, and customer requirements in addition to functionality.

My internship, career fairs, discussions with engineers and recruiters, team assignments, and Senior Design projects all motivated me to search for opportunities outside the classroom. They also encouraged me to be more at ease in settings where I do not have all the answers. Rather than shying away from something because I don't know enough about it, I now realize what it is that I need to learn.

Now having all my current knowledge, I would approach my studies in a much different way. If I was able to take some of the time that I used on simply completing assignments and preparing for the upcoming tests to learn the reasons and significance of the topics and how they relate to the future classes and engineering practice, then it would be much more beneficial for me. I also need to develop better studying skills early and raise questions right away when I do not understand something. There were several occasions when I did not use the existing resources, trying to complete my tasks by myself, which was rather counterproductive. Now I understand that it is perfectly fine and even necessary to address my questions to professors, classmates, teaching assistants, or experienced engineers to learn properly. I also wish that I had explored all the branches of electrical engineering much earlier.

My learning process has been modified due to these experiences. When dealing with complex engineering concepts, I am now able to dissect problems into smaller parts and ensure that I understand what is behind each equation. I use comparisons, solve problems, and apply simulations or engineering software in order to better understand the phenomenon at hand. Different software, such as MATLAB, Simulink, Cadence Virtuoso, LTspice, PLECS, ADS, and many others demonstrate various ways of linking theory to the behavior of the actual system. In addition, I have realized that each subject requires a different approach. Fabrication of semiconductors cannot be compared to power system problems, and circuit design does not equal embedded systems programming. The readiness to adjust my learning process according to different disciplines makes me more adaptable as a learner.

Recently, I have become fascinated by power systems as an example of my learning pattern. Indeed, power systems combine elements of circuits, mathematics, control theory, economics, and physical infrastructure. Whenever I encounter an unknown element, I begin studying the theory behind it and continue with examples, until I am able to link equations to the actual system. After that, I study the application of the concept in practice in actual power grids. The internship was extremely helpful here since I had the opportunity not only to study the theoretical aspects of electrical systems, but to see protection and test equipment in practice as well. At the same time, my experience with VLSI and circuit design provides me with another perspective on electrical engineering, which is applicable to a completely different scale.

With my undergraduate coursework drawing to a close, I am well aware that graduation will not mean the end of my education. My knowledge of technology, power systems, semiconductor devices, design packages, standards, and engineering practices will continue to evolve during my entire career. While I intend to build on my knowledge of power systems, it is also important for

me to retain the fundamentals of circuits and electronics which I have learned through my other courses. One of the things that I plan to do to develop further as an engineer is to study and prepare for the FE Electrical and Computer exam. In addition to that, I also plan to get experience with some of the common software and drawings used by engineers.

There have been many difficulties and challenges in my college experience at Iowa State University, but these challenges made my educational journey very enriching and productive. It was a difficult curriculum, challenging new concepts, projects that were not successful, and situations that required me to find a different approach to solving a certain problem. Difficulties such as these have developed in me resilience and flexibility. Initially, I entered the program in Electrical Engineering to acquire technical skills. However, I will leave this program with a completely different understanding of what being an engineer involves. Not only are technical skills important, but it is equally vital to be an effective communicator, work as part of a team, show ethical responsibility, have an inquisitive mind, and have the willingness to learn continuously. The biggest realization that I have gained from Iowa State is that my engineering education is not complete once I receive my degree.